

(1001CJA621620053)

Test Pattern

**CLASSROOM CONTACT PROGRAMME**

(Academic Session : 2020 - 2021)

JEE(Advanced)
REVIEW TEST
25-04-2021
JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : ALL STAR BATCHES)**ANSWER KEY****PAPER-1****PART-1 : PHYSICS**

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A	D	C	A	C	B	C	A,C	A,B,C,D	A,B,C,D	A,C
	Q.	11	12								
	A	A,D	A,B,C,D								
SECTION-II	Q.	1	2	3	4	5	6	7	8		
	A	1.33 to 1.34	1.25	4.00	2.66 to 2.67	0.06	0.00	2.00	13.33 to 13.34		

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A	B	A	D	B	B	B	A,B,C,D	A,B,C,D	A,B,C	A,B,C,D
	Q.	11	12								
	A	B,C,D	B,C,D								
SECTION-II	Q.	1	2	3	4	5	6	7	8		
	A	1.75	1.33	0.25	4.50	1.50	1.75	0.80	3.50		

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A	C	A	A	A	D	B	A,B,C,D	A	B,D	A,C
	Q.	11	12								
	A	A,B,C	A,B,C								
SECTION-II	Q.	1	2	3	4	5	6	7	8		
	A	4.00	2.00	0.31 or 0.32	2.50	364.00	39.06	2.50	1.00		

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JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : ALL STAR BATCHES)**ANSWER KEY****PAPER-2****PART-1 : PHYSICS**

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A	B,C	A,D	A,B,C,D	B,C	A,C,D	A,B,C,D	A,B,C,D	A,B,C,D	A	B
	Q.	11	12								
	A	C	D								
SECTION-II	Q.	1	2	3	4	5	6	7	8		
	A	0.25	15.00	2.50	4.30 to 4.40	2.00	3.00	3.00	7.00		

PART-2 : CHEMISTRY

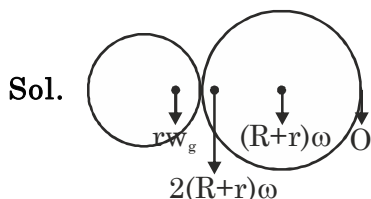
SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A	A,B,C,D	A,B,C	A,B,C,D	A,B	A,C,D	A,B,C	B,C	A,B,C	B	A
	Q.	11	12								
	A	C	D								
SECTION-II	Q.	1	2	3	4	5	6	7	8		
	A	0.01	0.30	1.52	3.50	1.25	3.00	8.70	9.74		

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A	A,B,C	A,B,C	B,C,D	D	A,B,C	A,C,D	B,C,D	A,B,C	A	B
	Q.	11	12								
	A	C	B								
SECTION-II	Q.	1	2	3	4	5	6	7	8		
	A	3.00	17.32 or 34.64	0.80	0.30	16.00	0.20	3.00	29.00		

PAPER-1**PART-1 : PHYSICS****SOLUTION****SECTION-I**

1. Ans. (D)



$$\frac{\omega_3}{\omega} = \frac{2(R+r)}{r} = 6$$

2. Ans. (C)

Sol. As no external force is acting on the given system of two particles, their centre of mass moves with a constant velocity. As initially both the particles are at rest their centre of mass is at rest. Hence, centre of mass is not moving. So we can say, they will meet at their location of centre of mass.

$$\therefore r_{cm} = \frac{4m \times 10m}{m + 4m} = 8m \text{ from A}$$

Since external force is zero, momentum is conserved. Hence,

$$m_A v_A + m_B v_B = 0$$

$$\therefore m_A v_A = -m_B v_B$$

3. Ans. (A)

Sol. $\int_0^{\pi/2} [\rho g (H - R \cos \theta) \{2\pi (R \sin \theta) (R d \theta)\}] \cos \theta = Mg$

$$H = \frac{2}{3} R + \frac{M}{\pi \rho_B R^2}$$

4. Ans. (C)

Sol. $F = -\frac{du}{dx} = \frac{\alpha}{x^3} = -kx$

for equilibrium

$$F = 0$$

$$\frac{\alpha}{x_3^0} = kx_0$$

$$\alpha = kx_0^4$$

$$F = \frac{\alpha}{(x_0 + x')^3} - k(x_0 + x')$$

$$= \frac{\alpha}{x_0^3} \left(1 - \frac{3x'}{x_0}\right) - k(x_0 + x')$$

$$= \frac{-3\alpha}{x_0^4} x' - kx'$$

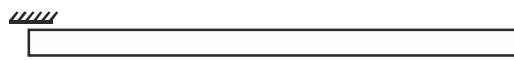
$$= \frac{-3(kx_0^4)}{x_0^4} x' - kx'$$

$$= -4kx'$$

5. Ans. (B)

Sol. $\alpha = \frac{mg \frac{\ell}{2}}{\frac{m \ell^2}{3}} = \frac{3g}{2\ell}$

$$\text{for } x \geq \frac{21}{3}$$



$$a \geq g$$

6. Ans. (C)

7. Ans. (A,C)

Sol. Minima will be formed at O,

$$\text{if } SS_1 + S_1O - SS_2 - S_2O = \frac{n\lambda}{2}, n = 1, 2, \dots$$

For minimum value of d, n = 1

$$\therefore \frac{\lambda}{4} = \sqrt{1 + d^2} - 1$$

$$(1 + d^2)^{1/2} - 1 = \frac{\lambda}{4}$$

$$1 + \frac{d^2}{2} - 1 = \frac{\lambda}{4} \quad (\text{Neglecting smaller terms})$$

$$\Rightarrow d + \sqrt{\frac{\lambda}{2}}; \beta = \frac{D\lambda}{d} = \sqrt{2\lambda}$$

8. Ans. (A,B,C,D)

Sol. for i to be maximum

$$\frac{di}{dt} = 0$$

$$\Rightarrow V_L = 0$$

$$(C) \Delta Q = \Delta u + \Delta \omega$$

$$0 = \Delta u + \frac{1}{2} kx^2$$

9. Ans. (A,B,C,D)

Sol. When $i = \max \Rightarrow \frac{di}{dt} = 0 \Rightarrow V_L = 0$

Let final potential of C_1 be V

$$C_1 V + C_2 V = C_1 V_1 - C_2 V_2$$

$$V = \frac{C_1 V_1 - C_2 V_2}{C_1 + C_2}$$

10. Ans. (A,C)

11. Ans. (A,D)

Sol. $Bev = \frac{mv^2}{r}$

$$\therefore B = \frac{mv}{er} = \frac{9 \times 10^{-31} \times 10^7}{1.6 \times 10^{-19} \times (5 \times 10^{-2})}$$

$$= 1.125 \times 10^{-3} T = 1.13 \text{ mT}$$

$$\omega = \frac{eB}{m}$$

$$\therefore T = \frac{2\pi m}{Be}$$

$$= \frac{2\pi \times 9 \times 10^{-31}}{1.13 \times 10^{-3} \times 1.6 \times 10^{-19}} = 3.14 \times 10^{-8} \text{ s}$$

Time taken to travel from A to B

$$= \frac{T}{2} = 1.57 \times 10^{-8} \text{ s}$$

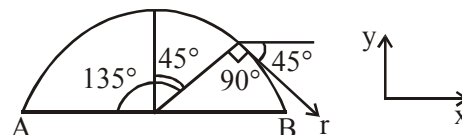
When $t = 0.785 \times 10^{-8} \text{ sec}$, it corresponds to $t = T/4$

Hence at $t = T/4$, the velocity vector is parallel to x-axis.

Hence angular momentum = mvr

$$= 9.1 \times 10^{-31} \times 10^7 \times 5 \times 10^{-2}$$

$$= 45.5 \times 10^{-26} = 4.5 \times 10^{-25} \text{ kg-m}^2\text{s}^{-1}$$



At $t = \frac{3T}{8}$ which corresponds to 135° as shown in Figure, the velocity vector makes an angle of $\frac{\pi}{4}$ in the clockwise direction,

i.e., $-\frac{\pi}{4}$ (by convention).

12. Ans. (A,B,C,D)

SECTION-II

1. Ans. 1.33 to 1.34

Sol. $\Rightarrow E.(2\pi R) = -\alpha\pi R^2$

$$E = -\frac{\alpha R}{2}$$

$$\therefore v = \frac{ft}{m} = \frac{\alpha q R t}{2m}$$

$$\Rightarrow F - qvB = \frac{mv^2}{R}$$

$$F = \frac{\alpha q^2 R t}{4m} (2B_0 + \alpha t)$$

2. Ans. 1.25

Sol. $t = \frac{d}{v}$ and $v = \sqrt{\frac{2eV}{m}}$ (2')

When the electron enters the disk, the impulse is

$$m\Delta v = t(evB) \quad (2')$$

$$\Rightarrow \Delta v = \frac{evBt}{m} = \frac{eBd}{m}$$

$$\Rightarrow \frac{\Delta v}{v} = \frac{eBd}{mv} = \frac{r}{L} \quad \text{where } r \leq R \quad (3')$$

$$\Rightarrow B = \frac{r}{dL} \sqrt{\frac{2mV}{e}} \quad (1')$$

3. Ans. 4.00

Sol. $\int \vec{E}_B \cdot \vec{E}_B dV = \frac{2}{\epsilon_0} \int \frac{1}{2} \epsilon_0 \vec{E}_B \cdot \vec{E}_B dV$

$$= \frac{2}{\epsilon_0} [\text{self energy of } q_B]$$

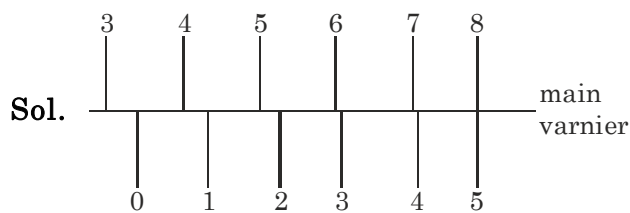
$$= \frac{2}{\epsilon_0} \left[\frac{1}{2} \frac{kq^2}{R} \right] = \frac{1}{\epsilon_0} \frac{kq^2}{R}$$

$\int \vec{E}_A \cdot \vec{E}_B dV = \frac{1}{\epsilon_0} \int \epsilon_0 \vec{E}_A \cdot \vec{E}_B dV$

$$= \frac{1}{\epsilon_0} [\text{Interaction energy of } q_A \text{ and } q_B]$$

$$= \frac{1}{\epsilon_0} \left[\frac{kq^2}{4R} \right]$$

4. Ans. 2.66 to 2.67



5. Ans. 0.06

Sol. $\tau = (I\ell B) \frac{\ell}{2}$

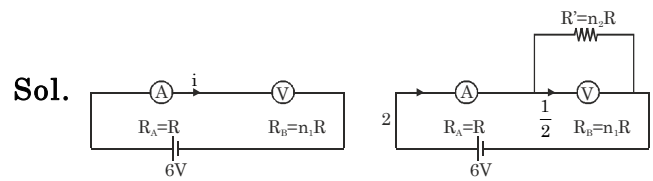
$$= 6 \times \frac{2}{10} \times \frac{1}{2} \times \frac{1}{10}$$

6. Ans. 0.00

Ans. $\frac{I_1}{I_2} = \frac{2\pi^2 \rho V n_1^2 A_1^2}{2\pi^2 \rho V n_2^2 A_2^2} = \frac{T_2^2 A_1^2}{T_1^2 A_2^2} = \frac{8^2 \times 3^2}{1^2 \times 2^2} = 1$

$$\beta_1 - \beta_2 = 1 - \log_{10} \frac{I_1}{I_2} = 0$$

7. Ans. 2.00



$$\frac{n_2 R}{n_1 R + n_2 R} (2i) = \frac{1}{2} \Rightarrow n_1 = 3n_2$$

$$R_{eq} = 2R_{eq}$$

$$R(1+n) = 2 \left[R + \frac{n_1 R n_2 R}{n_1 R + n_2 R} \right] \Rightarrow n_2 = \frac{2}{3}$$

$$\Rightarrow n_1 = 2$$

8. Ans. 13.33 to 13.34

Sol. Initially object is virtual

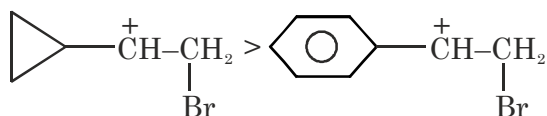
PART-2 : CHEMISTRY

SOLUTION

SECTION-I

1. Ans. (B)

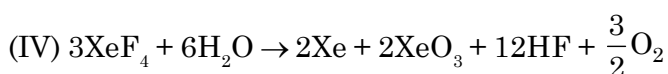
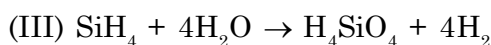
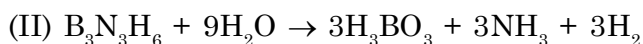
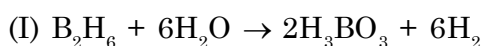
Sol. Stability of intermediate follows the order



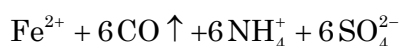
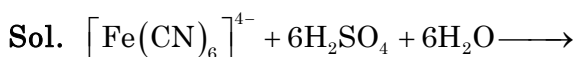
2. Ans. (A)

3. Ans. (D)

Sol.



4. Ans. (B)



5. Ans. (B)

6. Ans. (B)

7. Ans. (A, B, C, D)

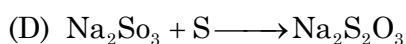
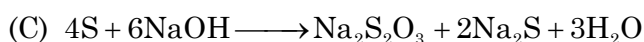
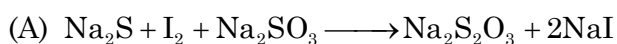
Sol: (A) Salt of meta-acids is polymeric

(B) Boric acid is polymeric solid due to extensive H-bonding

(C) BeH_2 and BeCl_2 are polymeric covalent solid

8. Ans. (A, B, C, D)

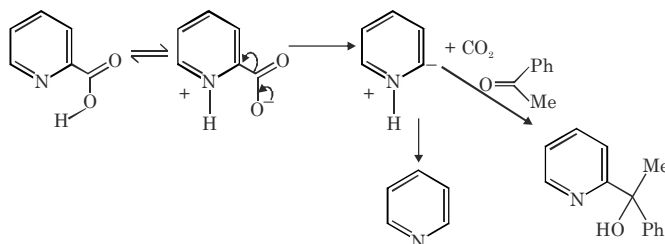
Sol.



9. Ans. (A, B, C)

10. Ans. (A, B, C, D)

Sol.

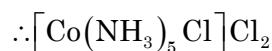


11. Ans: (B, C, D)

Sol: $\Delta T_f = i \cdot K_f \cdot m$

$$0.558 = i \times 1.86 \times 0.1$$

$$i = 3$$



12. Ans. (B, C, D)

SECTION-II

1. Ans. 1.75

Sol. Radial distribution

$$f_n = |R^2| r^2 \Psi_{3s}^2 r^2 = \frac{r^2}{(81)^2 (3\pi) a_0^3} \left(27 - \frac{18r}{a_0} + \frac{2r^2}{a_0^2} \right)^2 e^{-\frac{2r}{3a_0}}$$

At nodes, radial distribution function = 0

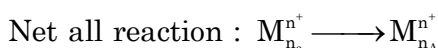
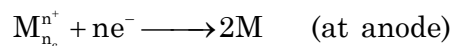
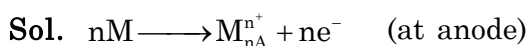
$$\therefore 27 - \frac{18r}{a_0} + \frac{2r^2}{a_0^2} = 0$$

$$\Rightarrow 2r^2 - 18ra_0 + 27a_0^2 = 0$$

$$\therefore r = 1.9a_0 \text{ and } 7a_0$$

$$\therefore x = 7$$

2. Ans. 1.33



$$\therefore 0.0295 = \frac{0.059}{n} \log \frac{[\text{M}_{n\text{c}}^{n+}]}{[\text{M}_{n\text{A}}^{n+}]}$$

$$\therefore n = 2$$

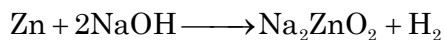
3. Ans. 0.25

Sol. Equivalents of X = Eq. of NH_3 = Equivalent of H_2SO_4 used = $10^{-2} \times 8$

$$\frac{2.616(2)}{A} = 10^{-2} \times 8$$

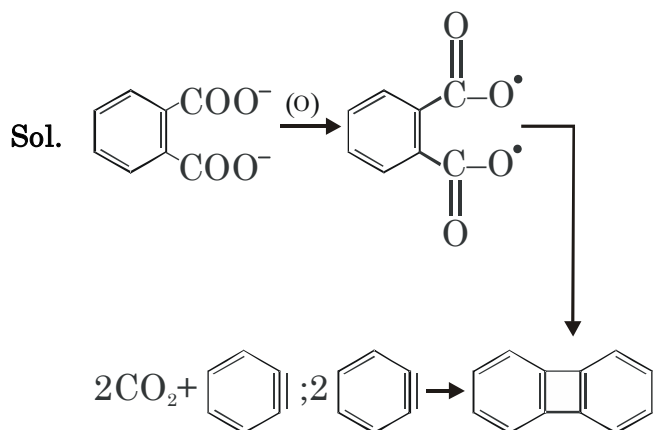
$$\therefore A = 65.4$$

'X' is Zn

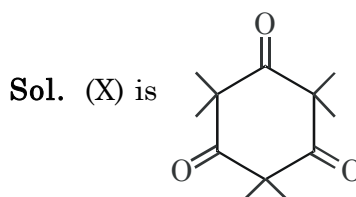


$\therefore$ 2 moles of H_2 gas would be liberated.

4. Ans. 4.50



5. Ans. 1.50



6. Ans. 1.75

7. Ans. 0.80

Sol. P_4S_{10} , Al_2S_3 , MgS and Cr_2S_3

8. Ans. 3.50

Sol. SnCl_2 , H_2O , NOCl , H_2S , I_3^+ , OF_2 , NH_2^-

PART-3 : MATHEMATICS

SOLUTION

SECTION-I

1. Ans. (C)

Sol. Let a , b , c be the intercepts cut off by the plane $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$ or $lx + my + nz = p$ on the coordinate axis.

We know $a = \frac{p}{l}$, $b = \frac{p}{m}$, $c = \frac{p}{n}$ where p is

length of perpendicular from origin to the plane and l , m , n are the direction cosines of the normals.

Foot of perpendicular from origin to the plane is $(\alpha, \beta, \gamma) = (pl, pm, pn)$

Now, $ab + bc + ca = 5$

$$\Rightarrow p^4 \left(\frac{1}{\alpha\beta} + \frac{1}{\beta\gamma} + \frac{1}{\gamma\alpha} \right) = 5$$

$$\Rightarrow (\alpha^2 + \beta^2 + \gamma^2)^2 \left(\frac{1}{\alpha\beta} + \frac{1}{\beta\gamma} + \frac{1}{\gamma\alpha} \right) = 5$$

$$\Rightarrow \text{Locus is } (x^2 + y^2 + z^2)^2 \left(\frac{1}{xy} + \frac{1}{yz} + \frac{1}{zx} \right) = 5.$$

2. Ans. (A)

Sol. $g(x)$ will be a quadratic polynomial with roots 1 and -1

$$\text{Let } g(x) = a(x-1)(x+1)$$

$$\therefore h(x) = \frac{(x-1)(x^3 - 2x^2 - 2x - 2)}{a(x-1)(x+1)}$$

$$\therefore \lim_{x \rightarrow 1} \frac{(x-1)(x^3 - 2x^2 - 2x - 2)}{a(x-1)(x+1)} = \frac{1}{2}$$

$$\Rightarrow -\frac{5}{2a} = \frac{1}{2} \Rightarrow a = -5$$

$$\therefore g(x) = -5(x^2 - 1)$$

$$\lim_{x \rightarrow 0} \frac{5(1 - \cos^2 x)}{5 \ln(1 + \sec^2 x - 1)} = \lim_{x \rightarrow 0} \frac{1 - \cos^2 x}{\sec^2 x - 1} = 1$$

3. Ans. (A)

$$\text{Sol. } 9A^{-1} = 4A^T \Rightarrow |9A^{-1}| = |4A^T|$$

$$\Rightarrow 9^5 \times \frac{1}{|A|} = 4^5 |A| \Rightarrow |A| = \left(\frac{3}{2} \right)^5$$

$$|\text{adj adj adj } A| = |A|^{4^3} = \left(\frac{3}{2} \right)^{320}$$

Number of digits in $|\text{adj adj adj } A|$

$$= \left[\log \left(\frac{3}{2} \right)^{320} \right] + 1 = 56.$$

([.] denotes greatest integer function).

4. Ans. (A)

Sol. Let $A\left(-\frac{5}{2}, \frac{\sqrt{7}}{2}\right)$, $B\left(\frac{3}{2}, \frac{\sqrt{7}}{2}\right)$ and

$P(a, 0)$. Then P is on x -axis for $d = PA - PB$.

$\therefore |PA - PB| < |AB| = 4$, and then $-4 < d < 4$ is established.

5. Ans. (D)

Sol. $\cos A = \frac{b^2 + c^2 - a^2}{2bc}$

$$\Rightarrow b^2 - (2c \cos A) b + c^2 - a^2 = 0$$

$$b_1 + b_2 = 2c \cos A \quad \dots(1)$$

$$b_1 b_2 = c^2 - a^2 \quad \dots(2)$$

using $b_1 = 2b_2$ in (1) & (2)

$$\text{we get } 2\left(\frac{2 \cos A}{3}\right)^2 = c^2 - a^2$$

$$8c^2 \cos^2 A = 9c^2 - 9a^2$$

$$8c^2(1 - \sin^2 A) = 9(c^2 - a^2)$$

$$\Rightarrow 3a = c\sqrt{1 + 8\sin^2 A}$$

6. Ans. (B)

Sol : $\frac{\sin^2 \theta \cdot \sin \theta}{\cos \theta} + \frac{\cos^3 \theta}{\sin \theta} = (1 + \sin 2\theta) + \frac{(\sin^2 \theta + \cos^2 \theta)}{\sin \theta \cdot \cos \theta}$

$$\Rightarrow \frac{\sin^4 \theta + \cos^4 \theta}{\sin \theta \cdot \cos \theta} = (1 + \sin 2\theta) + \frac{1}{\sin \theta \cdot \cos \theta}$$

$$\Rightarrow \frac{1 - 2\sin^2 \theta \cos^2 \theta}{\sin \theta \cdot \cos \theta} = (1 + \sin 2\theta) + \frac{2}{(\sin 2\theta)}$$

$$\Rightarrow \frac{2\left(1 - \frac{\sin^2 2\theta}{2}\right)}{\sin 2\theta} = (1 + \sin 2\theta) + \frac{2}{\sin 2\theta}$$

$$\Rightarrow \frac{2\left(1 - \frac{t^2}{2}\right)}{t} = (1 + t) + \frac{2}{t} \Rightarrow \frac{(2 - t^2)}{t} = \frac{t^2 + t + 2}{t}$$

$$\Rightarrow 2t^2 + t = 0$$

$$\Rightarrow t(2t + 1) = 0 \Rightarrow t = 0$$

$$\& \quad t = -\frac{1}{2}$$

$$\sin 2\theta = 0$$

$$\sin \theta = 0; \cos \theta = 0 \quad \sin 2\theta = -\frac{1}{2}$$

7. Ans: (A,B,C,D)

Sol. $\int_0^{\pi/2} \sin x \ln |\sin x - \cos x| dx$

$$= \int_0^{\pi/4} \ln |\sin x - \cos x| (\sin x + \cos x) dt$$

$$\text{Put, } \sin x - \cos x = t = \int_{-1}^0 \ln |t| dt = -1$$

$$\Rightarrow \cos \theta - 1 + \cos 2\theta = -1$$

$$\Rightarrow 2\cos \frac{3\theta}{2} \cos \frac{\theta}{2} = 0$$

$$\Rightarrow \frac{\theta}{2} = \frac{\pi}{2} \Rightarrow \theta = \pi$$

$$\Rightarrow z = -1$$

$$\Rightarrow \frac{3\theta}{2} = \frac{\pi}{2} \Rightarrow \theta = \frac{\pi}{3}$$

$$z = e^{\frac{\pi}{3}i} = \frac{1}{2} + \frac{i\sqrt{3}}{2}$$

$$\Rightarrow z + \bar{z} = 1$$

8. Ans : (A)

Sol : $g(x) = e^{-x} \int_0^x f(x) dx$

$$g'(x) = e^{-x} f(x) - e^{-x} \int_0^x f(x) dx = 0$$

$$g'(c) = 0 \Rightarrow \int_0^c f(x) dx = f(c)$$

9. Ans. (B,D)

Sol. C is $\frac{x^2}{7} - \frac{y^2}{3} = 21$ ($y \geq 0$)

$\therefore$ Equation of L will be $y = x - 2$ and

$$A\left(\frac{7}{2}, \frac{3}{2}\right)$$

Equation of circle

$$\left(x - \frac{7}{2}\right)^2 + \left(y - \frac{3}{2}\right)^2 + \lambda(x - y - 2) = 0$$

passing through (0, 0)

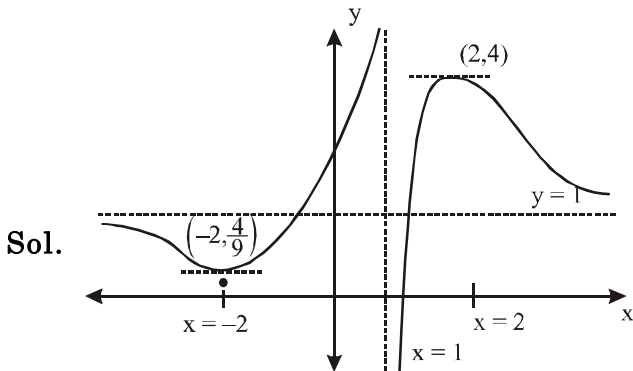
$$\Rightarrow \lambda = \frac{29}{4}$$

10. Ans : (A,C)

$$\text{Sol : } |a-b|^2 - |1-\bar{a}b|^2 = (|a|^2 + |b|^2 - 2\text{Re}(\bar{a}b)) - (1 + |a|^2|b|^2 - 2\text{Re}(\bar{a}b)) = -(1-|a|^2)(1-|b|^2) < 0$$

$$\text{Hence, } |a-b| < |1-\bar{a}b| \Rightarrow \left| \frac{a-b}{1-\bar{a}b} \right| < 1$$

11. Ans: (A,B,C)



Sol.

Graph of $f(x)$.

Now, verify alternatives.

12. Ans. (A,B,C)

Sol. Given

$$f(x) = e^x \int_0^x e^{-y} f'(y) dy - (x^2 - x + 1)e^x \dots (1)$$

$$\Rightarrow f'(x) = e^x \cdot e^{-x} f'(x) +$$

$$e^x \int_0^x e^{-y} f'(y) dy - (x^2 - x - 1)e^x - e^x(2x - 1)$$

$$\text{or, } e^x \int_0^x e^{-y} f'(y) dy = (x^2 - x + 1)e^x + e^x(2x - 1)$$

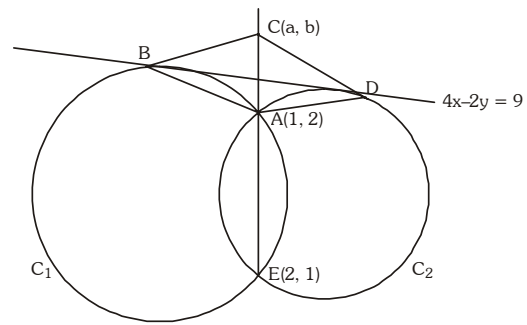
Now using (1)

$$f(x) + (x^2 - x + 1)e^x = (x^2 - x + 1)e^x + e^x(2x - 1) \\ \Rightarrow f(x) = e^x(2x - 1), \text{ Now verify alternatives.}$$

SECTION-II

1. Ans. 4.00

Sol. The radical axis bisectse the common tangent BD. Hence M is mid point of BD



Now C (a, b) lies on common chord AE which is

$$y - 2 = -(x - 1) \text{ or } x + y = 3$$

$$\therefore a + b = 3$$

$$\text{Also } M\left(\frac{a+1}{2}, \frac{b+2}{2}\right) \text{ lies on } 4x - 2y = 9$$

$$4\left(\frac{a+1}{2}\right) - 2\left(\frac{b+2}{2}\right) = 9$$

$$\Rightarrow 2a + 2 - b - 2 = 9 \Rightarrow 2a - b = 9 \dots (2)$$

2. Ans. 2.00

Sol. Clearly, point P lies on the plane passing through A,B and $\perp$ to xy-plane.

equation of plane is $x = 2y, z = 0$

$$P(\alpha, \beta) \text{ lies on plane } \Rightarrow \alpha = 2\beta$$

$$\therefore \left[\frac{\alpha + 3\beta}{5} \right] + \left[\frac{\alpha + 2}{\beta + 1} \right] = \left[\frac{5\beta}{5} \right] + \frac{2\beta + 2}{\beta + 1}$$

$$= [\beta] + 2 = 0 + 2 (\because 0 < \beta < 1)$$

$$\text{we can also find } \alpha = 4\sqrt{\frac{3}{5}} - 2 \text{ and } \beta = 2\sqrt{\frac{3}{5}} - 1$$

by transformation of axes.

3. Ans. 0.31 or 0.32

A_1 : $\rightarrow$ missing card is honour card

Sol. A_2 : $\rightarrow$ missing card is not honour card

B : $\rightarrow$ 1st 4 card drawn consist only one honour card

$$P\left(\frac{A_1}{B}\right) = \frac{P(A_1) \cdot P\left(\frac{B}{A_1}\right)}{P(A_1) \cdot P\left(\frac{B}{A_1}\right) + P(A_2) \cdot P\left(\frac{B}{A_2}\right)}$$

$$\begin{aligned} & \frac{4}{13} \cdot \frac{{}^{15}C_1 \times {}^{36}C_3}{{}^{51}C_4} \\ &= \frac{\frac{4}{13} \cdot \frac{{}^{15}C_1 \times {}^{36}C_3}{{}^{51}C_4} + \frac{9}{13} \times \frac{{}^{16}C_1 \times {}^{35}C_3}{{}^{51}C_4}}{\frac{4}{13} \cdot \frac{{}^{15}C_1 \times {}^{36}C_3}{{}^{51}C_4} + \frac{9}{13} \times \frac{{}^{16}C_1 \times {}^{35}C_3}{{}^{51}C_4}} = \frac{5}{16} \end{aligned}$$

4. Ans. 2.50

Sol. Let P denotes the chances of single bacteria to die, then

$$P = \frac{1}{4} + \frac{1}{2} \cdot P \cdot P + \frac{1}{4} P \cdot P \cdot P$$

$$\therefore P^3 + 2P^2 - 4P + 1 = 0 \Rightarrow (P - 1)(P^2 + 3P - 1) = 0$$

$$\therefore P = \frac{-3 + \sqrt{13}}{2}$$

$$\therefore 1 - P = 1 - \left(\frac{\sqrt{13} - 3}{2} \right) = \frac{5 - \sqrt{13}}{2}$$

5. Ans : 364.00

$$\begin{aligned} \text{Sol : } f(52, 14) &= f(14, 52) = \left(1 + \frac{14}{24} \right) \cdot f(14, 38) \\ &= \frac{26}{19} \left(1 + \frac{14}{24} \right) f(14, 24) = \frac{26}{19} \cdot \frac{19}{12} \left(1 + \frac{14}{10} \right) f(14, 10) = 364 \end{aligned}$$

6. Ans. (39.06)

Sol. 1 ball for box 2 and 1 for box 4 in ${}^7C_1 \times {}^6C_1 = 42$ ways

Remaining 5 balls are to be put in 3 boxes = $3 \times (2^5 - 2) + 3 \times 1 = 93$

$$\text{Total} = 42 \times 93 = 3906$$

7. Ans. (2.50)

$$\text{Sol. } x^2 \frac{f'(g(x))g'(x)}{f(g(x))} = (1 - 2x^2) \cdot \frac{g'f(x) \cdot f'(x)}{g(f(x))}$$

$$x^2 \frac{d}{dx} (\ln f(g(x))) = (1 - 2x^2) \frac{d}{dx} (\ln(gf(x)))$$

$$\int_0^k f(g(x)) dx = 1 - e^{-k^2}$$

$$f(g(k)) = 2k e^{-k^2}$$

$$\ln f(g(x)) = \ln(2x e^{-x^2})$$

$$\frac{d}{dx} \ln(f(g(x))) = \frac{2e^{-x^2} - 4x^2 e^{-x^2}}{2x e^{-x^2}} = \frac{2e^{-x^2}(-2x^2) + 1}{2e^{-x^2} \cdot x} = \frac{1}{x} - 2x$$

Now,

$$x^2 \frac{d}{dx} \ln(f(g(x))) = (1 - 2x^2) \frac{d}{dx} (\ln(gf(x)))$$

$$\Rightarrow x^2 \cdot \frac{1 - 2x^2}{x} = (1 - 2x^2) \frac{d}{dx} \ln(gf(x))$$

$$\Rightarrow \ln(gf(x)) = \frac{x^2}{x} + c$$

$$\Rightarrow \ln \left(\frac{gf(3)}{gf(2)} \right) = \frac{9}{2} - 2 = \frac{5}{2}$$

8. Ans. 1.00

$$\text{Sol. } 20^2 - 4c \geq 0$$

$$1 \leq c \leq 100$$

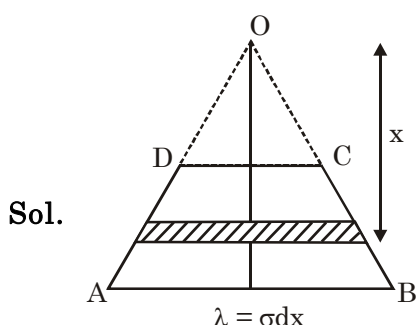
$$\text{Product} = 1 \cdot 2 \cdot 3 \dots 100 = 100!$$

$$n = 100$$

$$\text{Sum of digits} = 1$$

PART-1 : PHYSICS**SOLUTION****SECTION-I**

1. Ans. (B, C)



$$dE = 2 \left(\frac{1}{4\pi\epsilon_0} \right) \frac{\sigma dx}{x} \left(\frac{1}{2} \right)$$

$$dE = \frac{\sigma}{4\pi\epsilon_0} \frac{dx}{x}$$

$$E = \frac{7\sigma}{44\epsilon_0} \ln \sqrt{2}$$

2. Ans. (A, D)

Sol. Consider the bug at point P at any time t, moving with speed v along the stick, so $L = mvh = mvt \sin \theta \cos \theta \Rightarrow$ Angular momentum of bug about point O

$$\Rightarrow \tau = \frac{dL}{dt} = (m\ell \sin \theta \cos \theta) \frac{dv}{dt}$$

$$\Rightarrow mg(\ell - x) \cos \theta = (m\ell \sin \theta \cos \theta) \frac{dv}{dt}$$

$$\Rightarrow \frac{dv}{dt} = \frac{g}{\sin \theta} \left(1 - \frac{x}{\ell} \right)$$

$$\Rightarrow \frac{dv}{dt} + \frac{g}{\ell \sin \theta} x = \frac{g}{\sin \theta} \quad \text{Equation of SHM}$$

with mean position at point A

$$\Rightarrow T = 2\pi \sqrt{\frac{\ell \sin \theta}{g}}$$

$$\text{Time required} = \frac{T}{4} = \frac{\pi}{2} \sqrt{\frac{\ell \sin \theta}{g}}$$

3. Ans. (A, B, C, D)

4. Ans. (B, C)

$$\text{Sol. } E = m_1 \omega_1^2 A_1^2 = m_2 \omega_2^2 A_2^2$$

$$f = \frac{V}{2L} \Rightarrow \frac{f_1}{f_2} = \frac{\omega_1}{\omega_2} = \frac{L_2}{L_1} = \frac{4}{1}$$

$$\frac{A_1}{A_2} = \sqrt{\frac{m_2}{m_1} \times \left(\frac{\omega_2}{\omega_1} \right)^2}$$

$$= \sqrt{\frac{L_2}{L_1} \times \left(\frac{L_2}{L_1} \right)^2} = \frac{1}{2}$$

5. Ans. (A, C, D)

6. Ans. (A, B, C, D)

$$\text{Sol. (A) } \Delta K = \frac{1}{2} \mu v_{\text{rel}}^2$$

$$(B) a_{mM} = a_m - a_M = -F \frac{(M+m)}{mM}$$

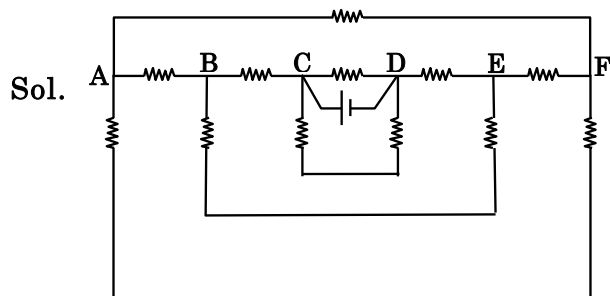
$$0^2 = u^2 - 2d \times \frac{F(M+m)}{mM} \Rightarrow F = \frac{mMu^2}{2d(M+m)}$$

7. Ans. (A, B, C, D)

Sol. Since only 6 different wavelength are excited, therefore highest excited state is $n = 4$.

Two wavelengths are shorter than λ_0 , initially atoms were in excited state $n = 2$
 Corresponding transitions are $4 \rightarrow 3$, $4 \rightarrow 2$, $4 \rightarrow 1$, $3 \rightarrow 2$, $3 \rightarrow 1$, $2 \rightarrow 1$.

8. Ans. (A, B, C, D)



use KVL

9. Ans. (A)

10. Ans. (B)

11. Ans. (C)

12. Ans. (D)

Sol. Centre of mass will free fall while particle as well as centre of box both perform uniform circular motion in the frame of cm

with angular speed $\omega = \frac{\sqrt{2g\ell}}{\ell} = \sqrt{\frac{2g}{\ell}}$

$$(A) T = \frac{\mu V_{rel}^2}{\ell} = \frac{m}{2} \frac{(2g\ell)}{\ell} = mg$$

$$(B) V = \sqrt{(gt)^2 + \left(\frac{\sqrt{2g\ell}}{2}\right)^2}$$

$$(C) \begin{array}{c} \text{Diagram of a circle with a vertical arrow pointing upwards from the center, labeled } a_{rel} = \frac{I}{m} = g \end{array}$$

$$(D) \begin{array}{c} \text{Diagram of a circle with a vertical arrow pointing upwards from the center, labeled } a_{rel} = \frac{I}{m} = g \end{array}$$

SECTION-II

1. Ans. 0.25

Sol. Initially

$$P_1 = P_0 + \frac{mg}{A} = \frac{5}{2} \frac{mg}{A}$$

Finally

$$P_2 = P_0 - \frac{Mg}{A} = \left(\frac{3}{2}m - M\right) \frac{g}{A}$$

$$P_1 \frac{V}{2} = P_2 V$$

2. Ans. 15.00

$$\text{Sol. } x = \frac{I_{C.M.}}{mR} = \frac{R}{2}$$

3. Ans. 2.50

$$\text{Sol. } \alpha_1 = \frac{p - p_1}{p_1(t_1 - t)} + \alpha$$

4. Ans. 4.30 to 4.40

Sol. 4.355×10^{-3} Joule

$$\frac{4}{3} \pi R^3 = N \frac{4}{3} \pi r^3 \quad \dots(i)$$

$$U_i = T4\pi R^2 \quad \therefore Q_L = U_i - U_f$$

$$U_f = T4\pi r^2 N$$

5. Ans. 2.00

$$\text{Sol. } T = \int_0^x \frac{\lambda_0 x}{L} g_{eff} dx = \frac{\lambda_0}{2L} x^2 g_{eff}$$

$$\therefore v = \sqrt{\frac{\lambda_0 x^2 g_{eff} L}{2L\lambda_0 x}} = \sqrt{15x}$$

$$\therefore \int_0^{15} \frac{dx}{\sqrt{15x}} = \int_9^t dt$$

$$\Rightarrow t = 2s$$

6. Ans. 3.00

Sol. Flux of magnetic field at any time t is

$$\phi = \int_t^{2\ell} \alpha x t \ell dx = \frac{3}{2} \alpha t \ell^3$$

$$\therefore |e| = \frac{3}{2} \alpha \ell^3$$

$$\text{charge on capacitor} = \frac{3}{2} C \alpha \ell^3$$

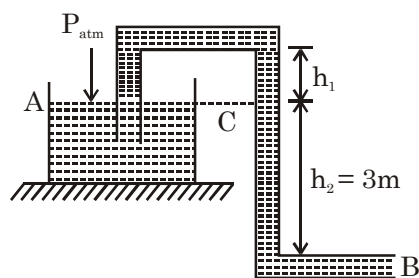
7. Ans. 3.00

$$\text{Sol. } \frac{dN}{dt} = -\lambda N + \frac{9\lambda N_0^2}{N} \Rightarrow N = 3N_0 \text{ if } t \rightarrow \infty$$

8. Ans. 7.00

Sol. $P_A = 1 \times 10^5 \text{ Nm}^{-2}$, $h_2 = 3 \text{ m}$.

Let ρ be the density of the water and P_{atm} be the atmospheric pressure at A and B openings. Applying Bernoulli's theorem for stream line motion,



$P_A = P_1 + \frac{1}{2}\rho v_1^2 + \rho g h_1$, where P_1 , v_1 are pressure and velocity at point C, at a height h_1 below the topmost point on the tube. Since the tube is of uniform cross-section and the liquid is incompressible, flow speed at C = flow speed at B i.e., $v_1 = v_B$.

From Torricelli's theorem between A and B.

$$v_B = \sqrt{2gh_2} = v_1$$

$$\therefore P_1 = P_A - \frac{1}{2}\rho \times 2gh_2 - \rho g h_1$$

$$= P_A - \rho g(h_1 + h_2)$$

Now pressure at C should be greater than or equal to zero (for negative pressure, water could not reach the top).

For maximum h_1 , P_1 should be minimum, equal to zero.

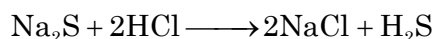
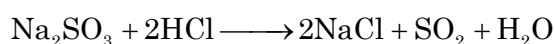
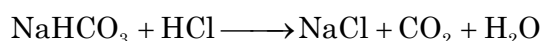
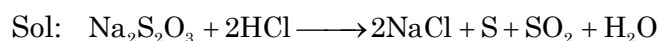
$$\therefore h_1 = \frac{P_A}{\rho g} - h_2 = \frac{1 \times 10^5}{10^3 \times 10} - 3 = 7\text{m}$$

PART-2 : CHEMISTRY

SOLUTION

SECTION-I

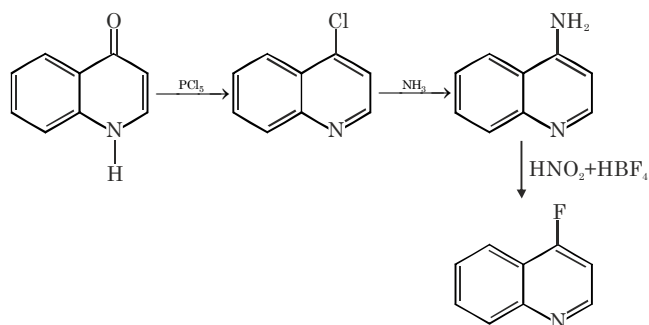
1. Ans.(A, B, C, D)



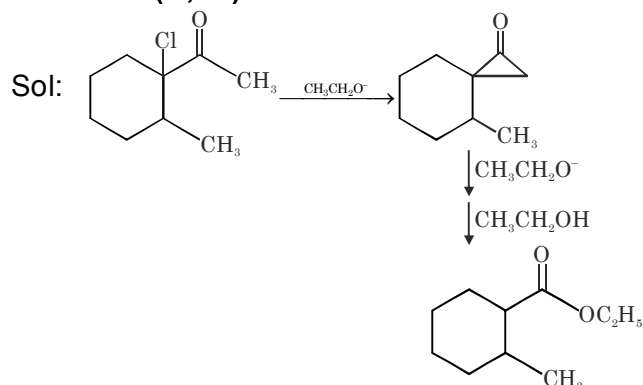
2. Ans.(A, B, C)

3. Ans.(A, B, C, D)

Sol:



4. Ans.(A, B)

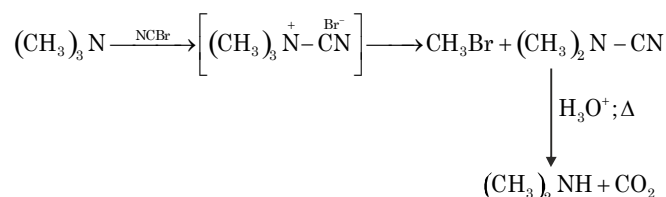


5. Ans.(A, C, D)

6. Ans.(A, B, C)

7. Ans.(B, C)

Sol:



8. Ans.(A, B, C)

Sol: Since the intermediate anion is planar, the stereochemistry of the starting substrate is irrelevant.

9. Ans.(B)

(Metal) (Methods of extraction)

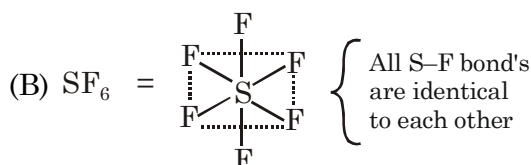
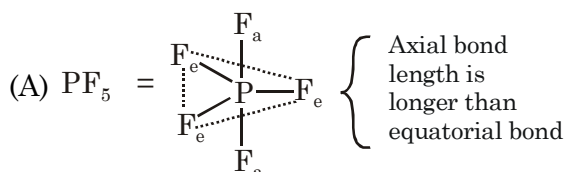
(A) Fe $\rightarrow$ by carbon reduction

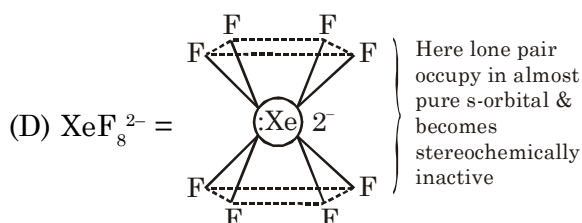
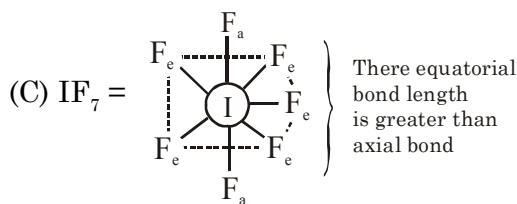
(B) Cu $\rightarrow$ by self reduction

(C) Mg $\rightarrow$ by electrolytic reduction

(D) Au $\rightarrow$ by cyanide process

10. Ans.(A)





11. Ans.(C)

12. Ans.(D)

SECTION-II

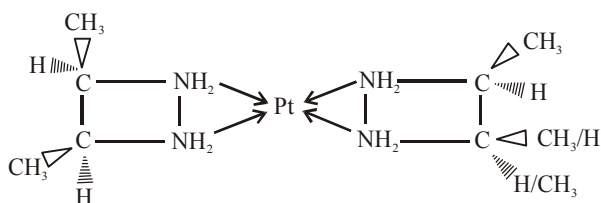
1. Ans.0.01

2. Ans.0.30

3. Ans.1.52

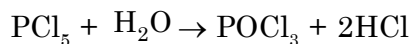
4. Ans.3.50

$[\text{Pt}(\text{bn})_2]^{2+}$ = Total isoemer is.



After changing CH_3 group from plane, we get total 5-G.I. and 2 G.I. are optically active isomers, hence total stereoisomers = 7

5. Ans.1.25



[Hence all 5-molecules produce POCl_3]

6. Ans.3.00

7. Ans.8.70

$$\text{Sol: } [\alpha]_D^{25} = \frac{\alpha}{\text{Cl}} = \frac{+1.74}{(1)(0.2)} = +8.7^\circ$$

8. Ans. 9.74

PART-3 : MATHEMATICS

SOLUTION

SECTION-I

1. Ans: (A,B,C)

$$\text{Sol. } \text{BD} = \text{DC} = \frac{a}{2}$$

$$\text{OD} = 2r_1$$

$$\Rightarrow \tan A = \frac{\text{BD}}{2r_1}$$

$$= \frac{a}{4r_1}$$

$$\frac{a^2}{r_1^2} + \frac{b^2}{r_2^2} + \frac{c^2}{r_3^2} = 16 (\tan^2 A + \tan^2 B + \tan^2 C) \geq$$

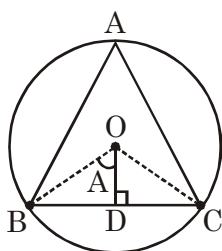
$$16 \times 9 = 144$$

2. Ans: (A,B,C)

$$\text{Sol. } \left| \int_0^a f(t) dt \right| = \left| a \int_0^1 f(ay) dy \right|$$

where $a \in [0, 1], y \in [0, 1]$

$$= \left| a \int_0^1 f(ay) dy - a \int_0^1 f(y) dy \right|$$



$$\text{because } \int_0^1 f(y) dy = 0$$

$$= a \left| \int_0^1 (f(ay) - f(y)) dy \right| \leq a \int_0^1 |f(ay) - f(y)| dy$$

.... (1)

using L.M.V.T. on $f(x)$, for $x \in [ay, y]$

$$\frac{f(ay) - f(y)}{ay - y} = f'(c), \text{ for some } c \in [ay, y]$$

$$\Rightarrow |f(ay) - f(y)| \leq 24(1 - a)y$$

$$\text{from (1), } \left| \int_0^a f(t) dt \right| \leq a \int_0^1 24(1 - a)y dy$$

$$= 12(1 - a)a \leq 3, \quad a \in [0, 1]$$

$$\Rightarrow \left| \int_0^a f(t) dt \right| \leq 3$$

Equality holds when $f(x) = 24x - 12$

3. Ans. (B, CD)

Sol. $AX = B$

$$\Rightarrow AX^2 = BX$$

$$\Rightarrow AX^2 = XB \quad (\because BX = XB)$$

$$\Rightarrow AX^2 = A$$

multiply both the side by A^{-1} ($\because A^{-1}$ exists)

$$\Rightarrow X^2 = I$$

$$\Rightarrow X^4 = I$$

$$= |X^4 + X^2 + I| = |3I| = 27$$

4. Ans: (D)

Sol. $(z + w)^2 = zw \Rightarrow z^2 + zw + w^2 = 0$

$$\text{Let } \frac{z}{w} = t \Rightarrow \frac{z}{w} = \frac{-1 \pm \sqrt{3}i}{2}$$

$$\arg z - \arg w = \frac{2\pi}{3} \text{ or } \arg z - \arg w = -\frac{2\pi}{3}$$

5. Ans. (A, B, C)

$$\text{Sol. } \int_2^n \frac{dx}{x} < a_n < \int_1^n \frac{dx}{x}$$

$$\Rightarrow \ln n - \ln 2 < a_n < \ln n \Rightarrow \lim_{n \rightarrow \infty} \left(\frac{a_n}{\ln n} \right) = 1$$

6. Ans. (A, C, D)

$$\text{Sol. } \begin{matrix} 2\alpha + \beta' = 5 \\ 2\alpha' + \beta = 5 \end{matrix} \Rightarrow \frac{\beta' - 2\alpha'}{\beta - 2\alpha} = 1$$

$$\Rightarrow \ln(\beta - 2\alpha) = x + c$$

put $x = 0$ we get $c = 0$

$$\beta - 2\alpha = e^x$$

$$5 - 2\alpha' - 2\alpha = e^x$$

$$\alpha' + \alpha = \frac{5}{2} - \frac{e^x}{2}$$

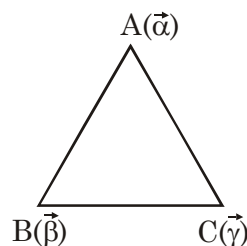
$$\therefore \alpha e^x = \int \left(\frac{5}{2} e^x - \frac{1}{2} e^{2x} \right) dx$$

$$\alpha = \frac{5}{2} - \frac{1}{4} e^x - \frac{5}{4} e^{-x}$$

$$\therefore \beta = 5 + \frac{e^x}{2} - \frac{5}{2} e^{-x}$$

7. Ans. (B, C, D)

$$\text{Sol. P.V. of } G = \frac{\vec{\alpha} + \vec{\beta} + \vec{\gamma}}{3}$$



Take P as origin

We have

$$|\vec{\alpha} + \vec{\beta} + \vec{\gamma}|^2 + |\vec{\alpha} - \vec{\beta}|^2 + |\vec{\beta} - \vec{\gamma}|^2 + |\vec{\gamma} - \vec{\alpha}|^2 = 3(|\vec{\alpha}|^2 + |\vec{\beta}|^2 + |\vec{\gamma}|^2)$$

$$\Rightarrow 9(PG)^2 + a^2 + b^2 + c^2 = 3((PA)^2 + (PB)^2 + (PC)^2)$$

$$\Rightarrow a^2 + b^2 + c^2 = 78 - 36 = 42$$

8. Ans: (A, B, C)

9. Ans. (A)

$$\text{Sol. (P)} \sum_{r=1}^n \frac{1}{r^2} - \frac{1}{(r+1)^2} = 1 - \frac{1}{(n+1)^2} = \frac{n(n+2)}{(n+1)^2}$$

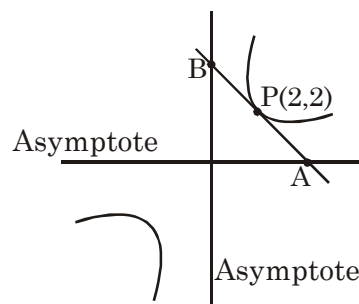
$$(Q) \lim_{n \rightarrow \infty} 1 - \frac{1}{(n+1)^2} = 1$$

$$(R) \prod_{k=1}^n \frac{k(k+2)}{(k+1)^2} = \frac{1.3}{2.2} \times \frac{2.4}{3.3} \times \frac{3.5}{4.4} \times \dots \times \frac{n(n+2)}{(n+1)(n+1)} = \frac{n+2}{2(n+1)}$$

$$(S) \lim_{n \rightarrow \infty} \frac{(n+2)}{2(n+1)} = \frac{1}{2}$$

10. Ans. (B)

Sol. $y + x = 4$ is tangent at (2, 2)



$$PB = PA = 2\sqrt{2}$$

As P is mid point of AB

Co-ordinates of A & B are given by

$$\frac{x-2}{\cos \theta} = \frac{y-2}{\sin \theta} = \pm 2\sqrt{2}$$

$$\text{Here } \cos \theta = -\frac{1}{\sqrt{2}}, \sin \theta = \frac{1}{\sqrt{2}}$$

$$\Rightarrow (x=0, y=4) \text{ \& } (x=4, y=0)$$

So joint equation of asymptotes

$$\Rightarrow xy = 0$$

$$\text{Equation of hyperbola } \Rightarrow xy = 4$$

11. **Ans. (C)**

Sol. 3, r_1, r_2, r_3 are inradius and three exradii of triangle formed by three lines

$$\frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3} = \frac{1}{r} = \frac{1}{3} \Rightarrow \frac{3}{\frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3}} = 9$$

and from given equation

$$r_1 r_2 + r_2 r_3 + r_3 r_1 = 243$$

$$\Rightarrow \frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3} = \frac{243}{r_1 r_2 r_3} \Rightarrow r_1 r_2 r_3 = 3^6$$

$$\Rightarrow (r_1 r_2 r_3)^{1/3} = 9 \Rightarrow r_1 = r_2 = 9$$

($\because$ HM = GM)

$$m = 27, n = 729$$

12. **Ans. (B)**

Sol. (P) $f(1) > f(2) > f(3) > f(4)$

then number of functions = ${}^{10}C_4$

(Q) $f(i) \neq f(j)$ for $i \neq j \Rightarrow$ number of one-one functions = ${}^{10}C_4 \cdot 4!$

(R) $f(1) \geq f(2) \geq f(3) \geq f(4)$, then number of functions is equal to number of non(-)ve integral solution of equation

$$x_1 + x_2 + x_3 + \dots \text{ upto } x_{10} = 4$$

$$\therefore \text{ number of solution } = {}^{13}C_4$$

(S) $f(1) < f(2) < f(3) > f(4)$,

$$\text{then number of function } = {}^{10}C_4 \times {}^3C_1$$

SECTION-II

1. **Ans 3.00**

Sol. Let P(h,k) be the point of intersection of the 2 tangents. Equation of tangent to $y^2 = 12x$

$$y - k = m(x - h)$$

$$\Rightarrow y = mx + k - mh$$

$$\text{where } k - mh = \frac{3}{m}$$

$$\Rightarrow hm^2 - km + 3 = 0 \dots (1)$$

Equation of tangent to $x^2 = 16y$

$$y - k = \left(\frac{-1}{m}\right)(x - h)$$

$$y = \left(\frac{-1}{m}\right)x + k + \frac{h}{m}$$

$$\text{Where } k + \frac{h}{m} = -4 \left(\frac{1}{m^2}\right)$$

$$\Rightarrow km^2 + hm = -4$$

$$\Rightarrow km^2 + hm + 4 = 0 \dots (2)$$

$$hm^2 - km + 3 = 0 \dots (1)$$

These 2 equations have a common root

$$\Rightarrow \frac{m^2}{3h+4k} = \frac{m}{4h-3k} = \frac{1}{-k^2-h^2}$$

$$\Rightarrow \frac{3h+4k}{-(k^2+h^2)} = \frac{(4h-3k)^2}{(k^2+h^2)^2}$$

$$\Rightarrow (3h+4k)(k^2+h^2) + (4h-3k)^2 = 0$$

$\Rightarrow$ Required locus is

$$(3x+4y)(x^2+y^2) + (4x-3y)^2 = 0$$

$$\Rightarrow \ell = 3, m = 4, n = 4$$

$$\Rightarrow \ell + n - m = 3$$

2. **Ans. 17.32 or 34.64**

Sol. 1, 4, 7, 10, 13, 16, $T_{2020} = 6058$

$$9, 16, 23, \dots t_{2020}$$

common terms are 16, 37, ...

Suppose there are n common terms

$$\Rightarrow 16 + (n-1)21 \leq 6058$$

$$\Rightarrow n-1 \leq \frac{6058-16}{21}$$

$$\Rightarrow n = 288$$

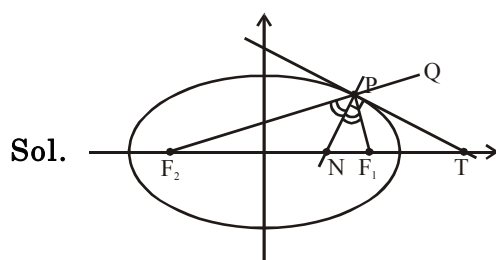
$\Rightarrow$ Number of uncommon terms in an A.P. = 1732

$\Rightarrow$ Number of uncommon terms in both A.P. = 3464

$$\Rightarrow \lambda = 1732 \text{ or } 3464$$

$$\Rightarrow \frac{\lambda}{100} = 17.32 \text{ or } 34.64$$

3. Ans. (0.80)



Sol.

As PN and PT are the angle bisectors of $\angle F_2PF_1$ & $\angle F_1PQ$ respectively (reflection property)

$$\Rightarrow \frac{|F_2P|}{|F_1P|} = \frac{|F_2N|}{|F_1N|} = \frac{|F_2T|}{|F_1T|}$$

Using componendo and dividendo

$$\left(\frac{|F_2N| + |F_1N|}{|F_2N| - |F_1N|} \right) \left(\frac{|F_2T| - |F_1T|}{|F_2T| + |F_1T|} \right) = 1$$

4. Ans. (0.30)

Sol. $f(x-1) = (x^2 - x + 1) \phi(x)$, where $f(x)$ is polynomial.

$$f(x) = (x^2 + x + 1) \phi(x+1)$$

$$\Rightarrow \phi(x) = \phi(x+1)$$

$\Rightarrow \phi(x)$ is a periodic function

$$\phi(x) = \text{constant}$$

$$f(x) = \lambda(x^2 + x + 1)$$

$$f(x) = (x^2 + x + 1)$$

5. Ans. (16.00)

Sol. $ABC = \text{diag. } (3 \cdot 11 \cdot 61, 2 \cdot 19 \cdot 53, 5 \cdot 13 \cdot 31)$

$$= \text{diag. } (2013, 2014, 2015)$$

$$\text{Trace of } ABC = 3 \times 2014 = 2 \times 3 \times 19 \times 53$$

$$\text{Numbr of factors} = 2^4 = 16$$

6. Ans. 0.20

Sol. Let $a + c + 2b = x$ (1)

$$a + b + 2c = y$$
 (2)

$$a + b + 3c = z$$
 (3)

$$c = z - y; b = x + z - 2y$$

$$a = -x + 5y - 3z$$

$$\left(\frac{2y-x}{x} \right) + \frac{4(x+z-2y)}{y} - \frac{8(z-y)}{z}$$

$$= -17 + 2 \left(\frac{y}{x} + \frac{2x}{y} \right) + 4 \left(\frac{z}{y} + \frac{2y}{z} \right)$$

$$= -17 + 4\sqrt{2} + 8\sqrt{2}$$

$$= 12\sqrt{2} - 17$$

7. Ans. 3.00

$$\text{Sol. } \overrightarrow{OG_1} \cdot \overrightarrow{BG_2} = 0 \Rightarrow \frac{\vec{a} + \vec{b} + \vec{c}}{3} \cdot \frac{\vec{a} + \vec{c} - 3\vec{b}}{3} = 0$$

$$\Rightarrow a^2 + c^2 - 3b^2 + 2\vec{a} \cdot \vec{c} - 2\vec{b} \cdot \vec{c} - 2\vec{a} \cdot \vec{b} = 0$$

Now,

$$|\vec{c} - \vec{a}|^2 = b^2, |\vec{c} - \vec{b}|^2 = a^2 \text{ and } |\vec{a} - \vec{b}|^2 = c^2$$

$$2\vec{a} \cdot \vec{c} = a^2 + c^2 - b^2, 2\vec{b} \cdot \vec{c} = b^2 + c^2 - a^2,$$

$$2\vec{a} \cdot \vec{b} = a^2 + b^2 - c^2$$

Putting in the above result, we get

$$2a^2 + 2c^2 - 6b^2 = 0 \Rightarrow \frac{a^2 + c^2}{b^2} = 3.$$

8. Ans. (29.00)

Sol.

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, X X X B B

$$n(S) = \frac{5!}{3!2!} = 10$$

B				
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, X X X B

$$n(E) = \frac{4!}{3!} = 4$$

$$P(E) = \frac{4}{10} = \frac{2}{5} = \frac{P}{Q}$$

$$\Rightarrow P^2 + Q^2 = 29$$